

Assignment No 21

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Domain : AI & ML

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Q1.

```
C: > Users > Administrator > app.py
1  # Import the Streamlit library to create the web application interface
2  import streamlit as st
3
4  # Import the pandas library to load and manipulate datasets
5  import pandas as pd
6
7  # Import the joblib library to load the trained machine learning model
8  # and other saved preprocessing files such as scaler and columns
9  import joblib
```

Q2.

```
11
12 import streamlit as st
13 import pandas as pd
14 import joblib
15 model = joblib.load("LR_model.pkl")
16 scaler = joblib.load("scaler.pkl")
17 encoded_columns = joblib.load("columns.pkl")
```

Q3.

```
20 import streamlit as st
21 import pandas as pd
22 import joblib
23
24     st.set_page_config(
25         page_title="Car Price Prediction app",
26         layout="centered"
27     )
28
```

Q4.

```
36 import streamlit as st
37 import pandas as pd
38 import joblib
39
40
41 # Display the main title of the application
42 st.title("Car Price Prediction app")
43
44 # Display a short description for the user
45 st.write("Enter the car details below to predict its sel
46
47
```

Q5.

```
8 year = st.number_input(  
9     "Manufacturing Year",  
10     min_value=1990,  
11     max_value=2025,  
12     value=2018  
13 )  
14  
15 # Mileage  
16 mileage = st.number_input(  
17     "Mileage",  
18     min_value=0,  
19     max_value=300000,  
20     value=20000  
21 )  
22
```

```
73 # Road Tax
74 tax = st.number_input(
75     "Road Tax",
76     min_value=0,
77     max_value=600,
78     value=150
79 )
80
81 # Miles Per Gallon (MPG)
82 mpg = st.number_input(
83     "MPG",
84     min_value=0.0,
85     max_value=150.0,
86     value=50.0
87 )
```

```
88  
89 # Engine Size  
90 engine_size = st.number_input(  
91     "Engine Size",  
92     min_value=0.0,  
93     max_value=8.0,  
94     value=1.5  
95 )
```

Q6.

```
108 # Transmission Type
109 transmission = st.selectbox(
110     "Transmission",
111     ["Automatic", "Manual", "Semi-Auto"]
112 )
113
114 # Fuel Type
115 fuel_type = st.selectbox(
116     "Fuel Type",
117     ["Petrol", "Diesel", "Hybrid", "Electric", "Other"]
118 )
```


Q7.

```
✓ car_model = st.text_input(  
    "Car Model",  
    placeholder="Enter car model name"  
)  
  
predict_button = st.button("Predict Price")
```

Q8.

```
if predict_button:  
    input_data = pd.DataFrame({  
        "year": [year],  
        "mileage": [mileage],  
        "tax": [tax],  
        "mpg": [mpg],  
        "engineSize": [engine_size],  
        "transmission": [transmission],  
        "fuelType": [fuel_type],  
        "model": [car_model]  
    })
```

Q9.

```
76 numerical_columns = ["year", "mileage", "tax", "mpg", "engineSize"]
77 |
78 input_data[numerical_columns] = scaler.transform(input_data[numerical_columns])
79
80 predicted_price = model.predict(input_data)
81
82 st.success(f"Predicted Car Price: £{predicted_price[0]:.2f}")
```

Q10.

Car Price Prediction App

Model

Fiesta

Year

2022

Mileage

5000

Tax

150

MPG

50.00

Engine Size

1.50

Transmission

Manual

Fuel Type

Petrol

Predict Price